PROJECT PROPOSAL

AlgoCred

Decentralized Academic Credentialing & Verification Ecosystem

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The Challenge

Use Case & Target Users

The Problem

Academic institutions face a **"Trust Deficit"** due to the global rise in forged physical and digital certificates. Current manual verification by HR departments takes weeks and is prone to human error.

Target Users

Universities & Colleges

To automate the issuance of tamper-proof, verifiable digital degrees.

Students

To maintain a lifelong, immutable portfolio of their achievements that they truly own.

Employers/Recruiters

To perform instant, zero-cost background checks without relying on third-party agencies.

How It Works

System Workflow

01

On-Chain Issuance

The Institution logs into a secure portal and initiates a transaction to mint a unique Soulbound NFT (SBT) for the student.

02

Decentralized Storage

The certificate's metadata (Name, GPA, Course) and asset (Image/PDF) are uploaded to IPFS via Pinata, ensuring the data is permanent and not stored on a single server.

03

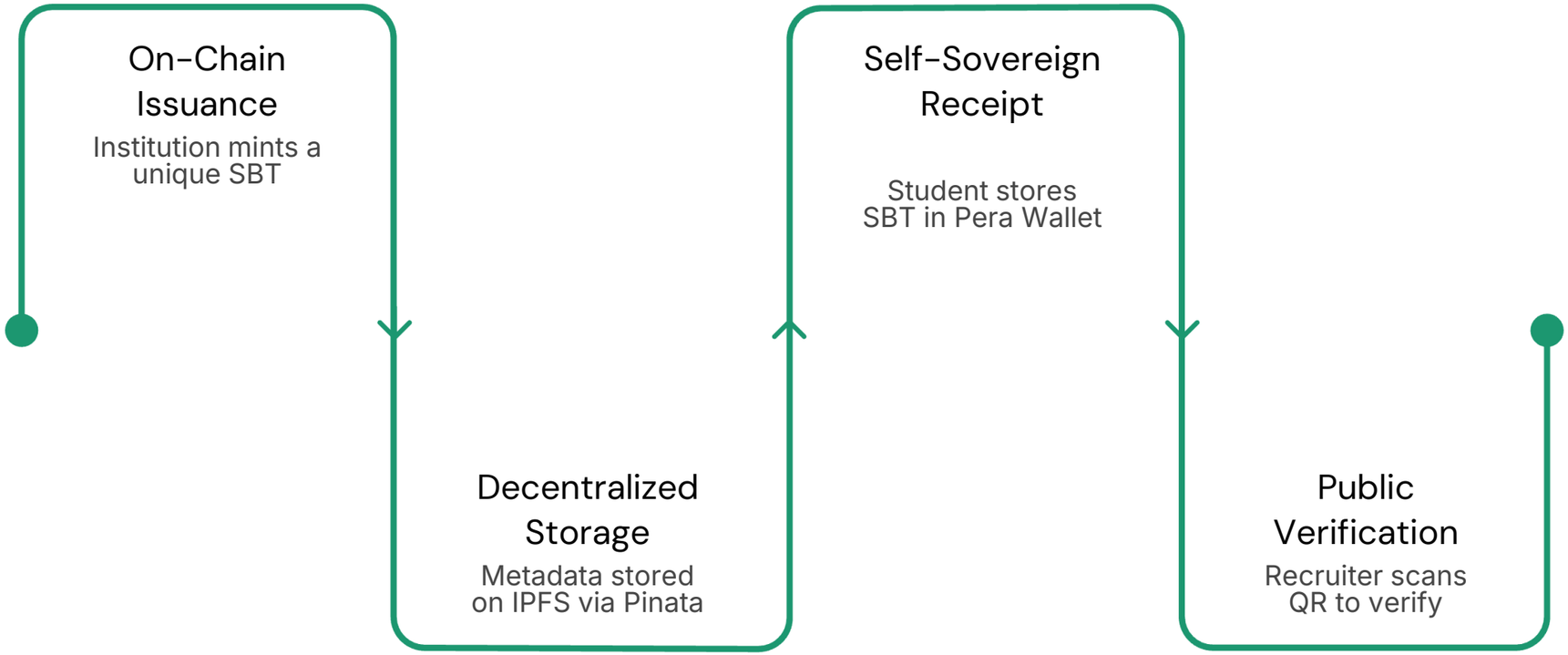
Self-Sovereign Receipt

The student receives the NFT in their Algorand wallet (Pera Wallet), where it is locked to prevent any transfer.

04

Public Verification

A recruiter scans a QR code on a student's resume, which queries the Algorand Indexer to verify the Asset's existence and the Institution's signature in real-time.



This end-to-end workflow ensures that every credential is verifiable, tamper-proof, and fully owned by the student from the moment of issuance.

Technical Architecture

Blockchain Layer

Algorand Layer-1 utilized for its 3.3s block finality and fork-proof consensus, ensuring credentials can never be duplicated or reverted.

Frontend Stack

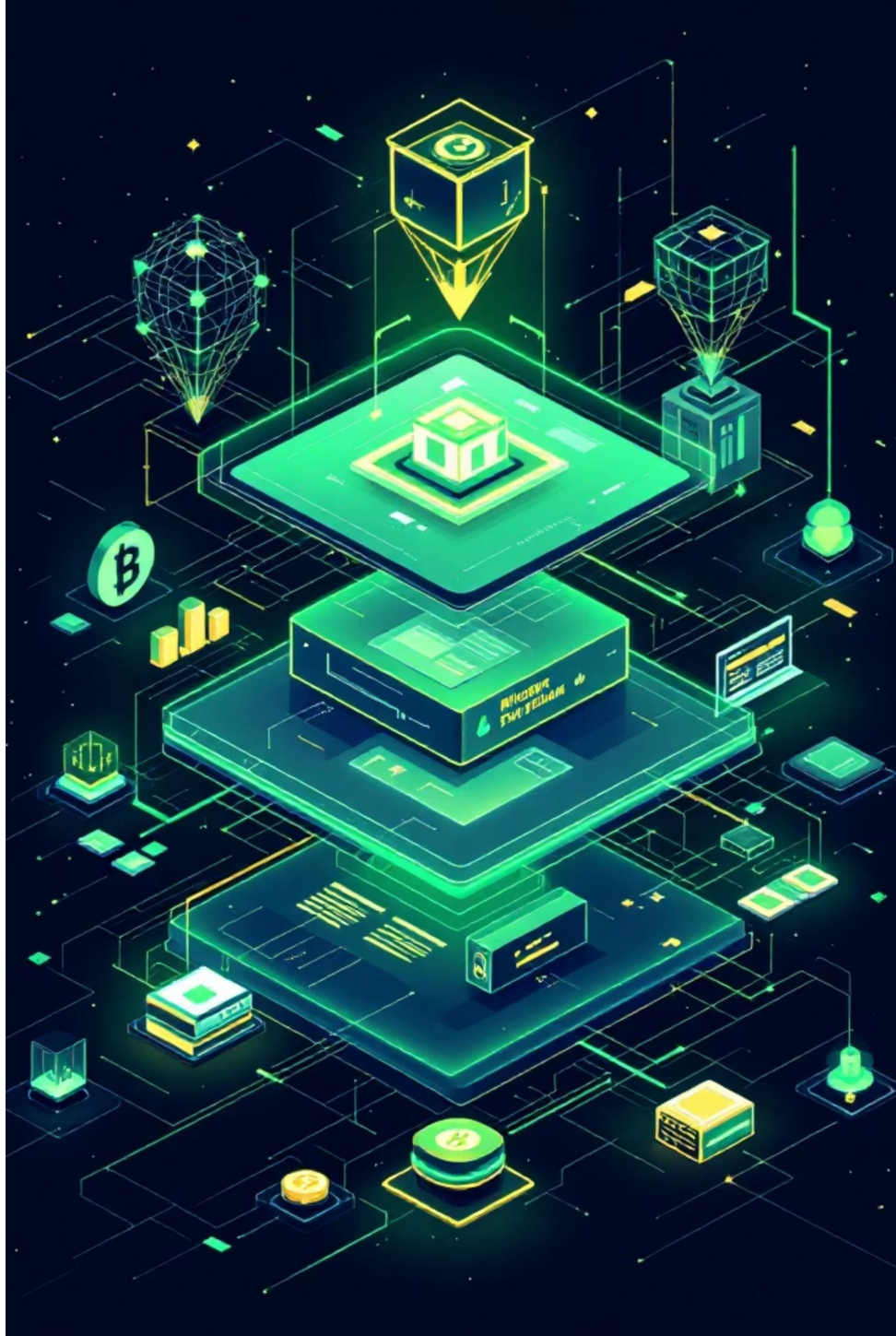
React.js & Tailwind CSS providing a dual-portal interface (Issuer & Student) with seamless Pera Wallet Connect integration.

Data Strategy

Use of Algorand SDK for atomic transactions and the Algorand Indexer for high-speed historical data fetching for verification.

Storage Layer

IPFS (InterPlanetary File System) to ensure the visual certificate is decentralized and always accessible.



Under the Hood

Smart Contract & Token Logic

1

ASA Configuration (NFT)

Each credential is a unique Algorand Standard Asset (ASA) with a total supply of 1 and 0 decimals.

2

Soulbound Logic (SBT)

We disable the student's ability to transfer the NFT by setting the Clawback and Freeze addresses to a controlled Institutional Address. This makes the certificate "Soulbound" to the student's identity.

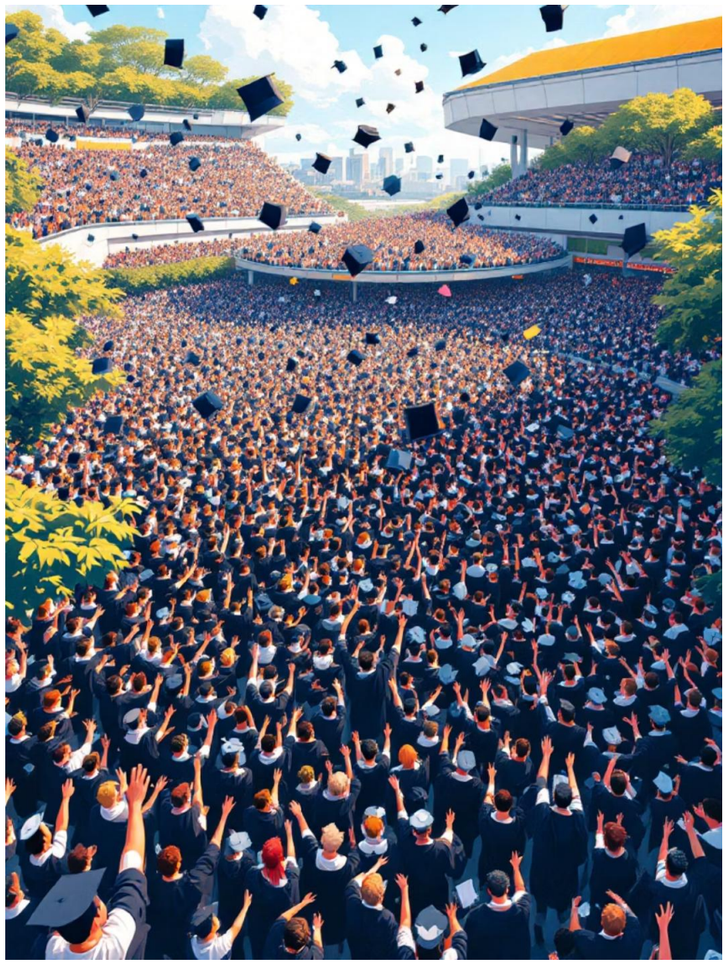
3

Verification Logic

A smart contract verification page that cross-references the CreatorAddress with a verified list of University public keys to ensure the NFT was minted by an authorized entity.

Why AlgoCred?

Feasibility, Scalability & Web3 Principles



Feasibility

Algorand's Pure Proof of Stake (PPoS) and minimal gas fees (0.001 ALGO) make it economically viable for mass adoption by state universities.

Scalability

The architecture supports Batch Minting using atomic groups, allowing thousands of graduates to receive degrees simultaneously within a single block.

Web3 Alignment

Built on the principle of Self-Sovereign Identity (SSI), ensuring students have full ownership of their data without a middleman.

The Winner's Edge

Innovation & Practicality

AI-Enhanced Resume Mapping

An AI-integrated layer that extracts skills from the verified on-chain metadata to build an auto-updating professional profile.

Zero-Knowledge Proofs (ZKP) Roadmap

Future implementation to allow students to prove they meet a specific GPA requirement (e.g., >7.5) without disclosing their exact score.

On-Chain Revocation

A mechanism to flag credentials as "Expired" or "Revoked" (e.g., in cases of misconduct) while keeping the original audit trail intact for transparency.